scientific reports

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Graph guided multiscale cross attention for multilabel chest X ray classification

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Multi-label chest X-ray (CXR) classification is challenging because thoracic abnormalities vary substantially in scale, visual saliency, and anatomical distribution, while disease labels often exhibit clinically meaningful dependencies. We propose a visual–semantic framework that integrates heterogeneous visual representations with graph-guided label reasoning for image-level multi-label CXR classification. The visual encoder consists of a Vision Transformer (ViT) branch and a DenseNet-121 branch with complementary inductive biases: the ViT branch provides self-attention-based content-adaptive token representations, whereas the DenseNet branch provides hierarchical convolutional feature maps with explicit spatial layouts. A multi-scale bidirectional dual cross-attention fusion (DCAF) module aligns these two representations and enables bidirectional cross-representation interaction at the C_3 and C_4 stages to construct a fused visual memory. To model label dependencies, we construct an ML-GCN-style label graph whose edges are derived from training-set conditional co-occurrence statistics and whose node features are initialized using GloVe label-name embeddings. The resulting GCN-refined label embeddings initialize the label queries of a Transformer decoder, which retrieves label-specific evidence from the fused visual memory and predicts a single logits matrix for multi-label classification. The proposed method achieves a Mean AUC of 0.849 on ChestX-ray14 following its official evaluation protocol and 0.815 on CheXpert using an internal 70%/10%/20% training/validation/testing partition. Qualitative Grad-CAM visualizations on selected cases further suggest that the proposed framework tends to produce activation patterns consistent with manually indicated visually suspicious regions; these visualizations are not intended as a formal localization evaluation. Overall, the results indicate that cross-representation visual fusion and graph-guided label-query decoding provide complementary benefits for multi-label CXR classification.

Chest X-ray (CXR) imaging is widely used for thoracic assessment because it is inexpensive, rapidly acquired, and broadly available in clinical practice. A single radiograph may contain multiple abnormalities, such as pleural effusion with atelectasis or cardiomegaly with pulmonary edema. Automated CXR interpretation is therefore naturally formulated as an image-level multi-label classification problem, in which several pathological findings must be predicted from the same image. Large-scale datasets such as ChestX-ray14 and CheXpert have accelerated the development of deep-learning systems for CXR analysis[1,2]. Nevertheless, reliable multi-label diagnosis remains difficult because abnormalities differ in lesion size, boundary clarity, radiographic appearance, and label frequency.

Convolutional neural networks (CNNs) have been widely used for CXR classification. For example, CheXNet uses DenseNet-121 to encode radiographic patterns through local filtering, weight sharing, and hierarchical downsampling[3]. Such convolutional inductive biases are useful for preserving spatially organized anatomical and textural structures. Vision Transformers (ViTs), by contrast, represent an image as a sequence of patch tokens and use self-attention to model content-adaptive interactions across distant image regions[4]. ViTs are therefore well suited for modeling long-range interactions among patch tokens, which can complement the spatial inductive biases of convolutional networks in CXR analysis. It should be noted that the distinction between CNNs and ViTs should not be interpreted as an absolutely rigid local–global separation. Deep convolutional layers can also encode broad contextual information as their receptive field expands, and ViT tokens still originate from local image patches. Nevertheless, CNNs and ViTs provide different inductive biases: DenseNet offers hierarchical convolutional features with explicit spatial layouts, whereas ViT provides self-attention-based token interactions

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for long-range relational modeling. In this work, we therefore view DenseNet and ViT as heterogeneous visual encoders with complementary representational properties rather than as strictly separated local and global branches.

Existing hybrid CNN–Transformer architectures often combine features through late concatenation, reweighting, or one-way attention[5,6]. These fusion strategies may be insufficient for CXR classification because disease evidence can appear at different spatial scales and may require interaction between spatially organized convolutional maps and self-attention-based token representations. To address this issue, we introduce a multi-scale bidirectional dual cross-attention fusion (DCAF) module. DCAF aligns DenseNet feature maps and ViT token features at the C_3 and C_4 stages and enables bidirectional cross-representation refinement before constructing the visual memory used by the decoder.

Label dependency is another important challenge in multi-label CXR classification. Many methods treat each disease as an independent binary target, although thoracic findings often co-occur or imply related clinical conditions. Graph-based approaches such as ML-GCN model label correlations through a graph built from co-occurrence statistics[7], while decoder-based methods such as Query2Label formulate multi-label prediction as label-query-based evidence retrieval[8]. However, learned label queries may lack explicit label-prior structure, and graph-based priors are often used as classifier parameters rather than being directly coupled with visual evidence retrieval. We address this limitation by constructing an ML-GCN-style directed label graph from training-set conditional co-occurrence statistics, initializing graph nodes with GloVe label-name embeddings, and using the GCN-refined label embeddings to initialize Transformer decoder queries. In this way, label-prior information participates directly in label-to-image evidence retrieval.

The key contributions of this work are summarized as follows:

- We propose a heterogeneous ViT–DenseNet visual encoder with a multi-scale bidirectional DCAF module. Unlike late concatenation or one-way fusion, DCAF aligns convolutional feature maps and Transformer token features at the C_3 and C_4 stages and enables bidirectional cross-representation refinement.
- We introduce a graph-guided label-aware Transformer decoder for multi-label CXR classification. The decoder queries are initialized by GCN-refined label embeddings obtained from an ML-GCN-style graph whose edges come from training-set conditional co-occurrence statistics and whose node features are initialized from GloVe label-name embeddings.
- We evaluate the proposed framework on ChestX-ray14 and CheXpert and analyze its major components through comparison studies, fine-grained ablations, multi-run averaged evaluation, computational profiling, healthy-sample analysis, and preliminary shared-label cross-dataset analysis. These experiments are designed to assess classification performance, the contribution of visual fusion and label-prior-guided decoding, and the behavior of the final model under complementary analysis settings.

Related work

CNN-based multi-label classification for chest X-rays

Public benchmarks enable systematic evaluation of multi-label CXR classification. ChestX-ray8/14 provide weak image-level labels and support joint prediction and coarse localization of common findings[1]. CheXpert further introduces uncertainty labels and an expert comparison protocol, which better reflects clinical reporting practice[2]. MIMIC-CXR aligns radiographs with free-text reports and supports subsequent progress in weakly supervised learning and representation pretraining[9]. Early multi-label CXR models were largely built on CNN backbones. CheXNet, using DenseNet-121, established a strong baseline and popularized the paradigm of independent sigmoid outputs for multi-label prediction[3,10]. Subsequent improvements commonly followed three directions. First, attention-based designs enhanced localization: AG-CNN introduces attention-guided refinement to highlight disease-relevant patterns, and CRAL learns category-wise attention to highlight discriminative patterns under comorbidity[11,12]. Second, structured clinical priors were introduced to reduce semantic inconsistency under noisy supervision, such as hierarchical disease dependencies and uncertainty-aware training[13]. Third, long-tailed label imbalance was addressed by asymmetric objectives and related designs that sharpen decision boundaries for rare findings[14,15]. Nevertheless, many CNN pipelines still rely on global pooling and independent sigmoid heads. This weakens long-range anatomical reasoning and leaves label dependency learning largely implicit, motivating hybrid encoders and explicit label-aware inference.

Vision transformers for chest X-ray classification

Vision Transformers (ViTs) use self-attention to perform content-adaptive token interaction, which makes them effective for modeling relationships among image patches in chest radiographs[4]. The Swin Transformer introduces a hierarchical shifted-window mechanism to improve computational efficiency while preserving multi-scale representation learning[16]. CXR-specific variants, such as IEViT, further adapt Transformer architectures to enhance sensitivity to subtle lesions and low-contrast textural patterns[17].

In medical imaging, the effectiveness of Transformer-based models is often shaped by both training-set scale and supervision quality. Park et al. use knowledge distillation to stabilize training and improve generalization in CXR diagnosis[18]. Self-supervised pretraining has also emerged as an effective strategy. CheXzero demonstrates that large-scale self-supervised learning can support expert-level pathology detection even without manual labels[19]. Masked autoencoder (MAE) and self-distillation frameworks, such as DINO and DINOv2, learn transferable representations for downstream tasks[20–22]. DINO-CXR further demonstrates improved multi-label recognition when annotated data are limited[23]. However, pure Transformer models may lack the explicit spatial continuity provided by convolutional feature maps, and their training can be less robust in the presence of label noise or limited data. These limitations motivate hybrid designs that combine CNN-style spatial inductive biases with Transformer-based content-adaptive token interactions.

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Multi-label decoding and cross-attention-based evidence retrieval

Conventional multi-label CXR classifiers often compress an image into a single pooled representation and predict labels with independent sigmoid heads. This can dilute localized evidence and provides limited mechanisms for consistent prediction under comorbidity. Transformer decoders provide a more structured alternative by maintaining label queries and retrieving relevant visual evidence through cross-attention[24]. Query2Label formulates multi-label classification as label-conditioned evidence retrieval by treating each label as a learnable query[8]. ML-Decoder scales this paradigm and supports semantic initialization with word vectors to improve generalization on long-tailed categories[25]. CXR-oriented decoders further extend this line: TransDD aggregates complementary visual evidence with a dual-path decoder, while Label Correlation Transformer explicitly models label relations and improves evidence aggregation[26,27]. Many existing decoder-based methods learn queries from random initialization or weak semantic priors and rely on single-scale visual memory. This can limit fine-grained retrieval and reduce robustness when label correlations shift across datasets. These limitations motivate the use of label-prior-guided queries together with multi-scale fused visual memory for evidence retrieval.

CNN–transformer hybrid encoding and multi-scale feature fusion

Hybrid encoders aim to combine convolutional inductive biases with attention-based token interaction. CoAtNet integrates convolution and attention across stages to improve generalization across data regimes[28]. MaxViT further enhances non-local token interaction via multi-axis attention while keeping computation manageable[29]. In CXR studies, two-branch architectures often compute CNN and Transformer features separately and fuse them only before prediction[5,6]. Many designs use concatenation or weighted summation, which limits how much one representation can correct the other during feature learning. This becomes critical when small lesions and widespread structural abnormalities must be recognized simultaneously. These observations motivate a fusion strategy that aligns heterogeneous visual features at multiple scales and uses bidirectional cross-attention to promote cross-representation refinement.

Label dependency modeling and prior-driven joint reasoning

Multi-label CXR prediction involves rich inter-label structure, including co-occurrence patterns, mutual exclusivity, and clinically meaningful complication pathways. Accordingly, graph-based approaches have been widely explored to model such dependencies. ML-GCN constructs a label graph from co-occurrence statistics and uses graph convolution to learn label embeddings, which are then used to parameterize interdependent classifiers[7]. Beyond label graphs, ImageGCN models inter-image relationships to improve representation consistency and disease identification[30]. More recent studies have explored correlation-guided label feature learning, unified correlation integration frameworks, and interpretable prediction through label embeddings[31–33]. In multi-label CXR classification, jointly modeling visual evidence and label co-occurrence priors has also been shown to benefit the analysis of complex cases[34].

Despite these advances, two limitations remain. First, label semantics and co-occurrence-derived graph structure are often incorporated in separate stages, making it difficult to couple label priors with visual evidence retrieval. Second, label priors are frequently used only for static classifier construction or post hoc recalibration, rather than being tightly integrated with evidence retrieval during inference. These limitations motivate an ML-GCN-style label graph whose edges are derived from training-set conditional co-occurrence statistics and whose node features are initialized from semantic label embeddings; the resulting label priors are injected directly into decoder queries.

Taken together, prior studies have advanced multi-label CXR classification through stronger CNN backbones, Transformer-based context modeling, hybrid visual encoders, decoder-based evidence retrieval, and graph-based label dependency modeling. However, unified frameworks that combine deep multi-scale visual fusion with label-prior-guided decoding remain underexplored. This gap motivates the visual–semantic framework proposed in this work, which couples a parallel ViT–DenseNet encoder with a graph-guided Transformer decoder for end-to-end integration of heterogeneous visual evidence and label priors.

Methods

Problem formulation

Given a mini-batch of CXR images $X \in \mathbb{R}^{B\times 3\times H\times W}$, the goal is to predict an image-level multi-label target matrix $Y \in \{0,1\}^{B\times L}$, where B is the batch size and L is the number of disease labels. The proposed model outputs a single logits matrix $Z \in \mathbb{R}^{B\times L}$, and the final multi-label probabilities are obtained by

$$\hat{Y} = \sigma(Z), \tag{1}$$

where $\sigma(\cdot)$ denotes the element-wise sigmoid function. This study focuses on image-level classification rather than lesion segmentation or detection; visualization results are therefore used only as qualitative evidence.

For clarity, we use the following notation throughout the method description. The mini-batch size is denoted by B, the input image height and width by H and W, and the number of disease labels by L. The DenseNet branch produces convolutional feature maps C_s at stages $s \in \{3,4\}$, where d_s, H_s, and W_s denote the channel dimension, height, and width of stage s, respectively. The ViT branch produces a token feature grid T after removing the class token, if present, where d_v, H_v, and W_v denote its channel dimension and spatial size. After projection and scale alignment, the corresponding features are denoted by C'_s and T_s. The superscript $\flat$ denotes flattening a two-dimensional feature map into a token sequence, and the tilde symbol denotes a cross-attention-refined feature. The superscript (s) in projection functions denotes scale-specific parameters rather than exponentiation. The DCAF module outputs scale-specific fused features F_s and an aggregated visual feature map F. The flattened visual memory is denoted by M. For the label graph, P denotes the directed conditional

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co-occurrence matrix, $\mathcal{B}$ denotes the thresholded binary matrix, and A denotes the ML-GCN-style re-weighted correlation matrix. The initial and final GCN label embeddings are denoted by $H^{(0)}$ and $H^{(2)}$, respectively.

Overview of the proposed framework

Figure 1 summarizes the proposed visual–semantic framework. The framework contains four main components: a heterogeneous ViT–DenseNet visual encoder, a multi-scale bidirectional dual cross-attention fusion module, an ML-GCN-style label graph, and a graph-guided Transformer decoder. The DenseNet-121 branch provides hierarchical convolutional feature maps with explicit spatial layouts, while the ViT branch provides self-attention-based token representations. These two branches are treated as heterogeneous visual encoders with complementary inductive biases rather than as rigidly separated spatial-range extractors.

The DCAF module aligns DenseNet feature maps and ViT token features at the C_3 and C_4 stages and performs bidirectional cross-representation interaction. The fused feature map is flattened into a visual memory M, which is later used by the Transformer decoder. On the label side, an ML-GCN-style directed label graph is constructed from training-set conditional co-occurrence statistics. The graph nodes are initialized with GloVe 840B label-name embeddings, and a two-layer GCN produces label embeddings used to initialize decoder queries. The decoder retrieves label-specific evidence from the fused visual memory and outputs the final logits Z.

Heterogeneous visual encoder

DenseNet-121 branch

The DenseNet-121 branch extracts hierarchical convolutional feature maps from intermediate stages. We denote the selected feature maps by

$$C_s \in \mathbb{R}^{B\times d_s\times H_s\times W_s}, \qquad s \in \{3,4\}. \tag{2}$$

Here, d_s, H_s, and W_s denote the channel dimension, height, and width of the DenseNet feature map at stage s, respectively. The C_3 feature map preserves relatively richer spatial structure after sufficient nonlinear transformation, whereas C_4 provides stronger semantic abstraction at a lower spatial resolution. These stages are selected because they offer complementary levels of spatial and semantic abstraction for image-level CXR classification.

ViT branch

The ViT branch divides the input image into patch tokens and encodes them with self-attention. After removing the class token when present, the remaining tokens are reshaped into a two-dimensional feature grid,

$$T \in \mathbb{R}^{B\times d_v\times H_v\times W_v}. \tag{3}$$

Here, d_v, H_v, and W_v denote the channel dimension, height, and width of the reshaped ViT token grid, respectively. The ViT representation provides content-adaptive token interactions. When MAE-pretrained weights are used, the pretraining is regarded as an initialization strategy that may improve patch-level

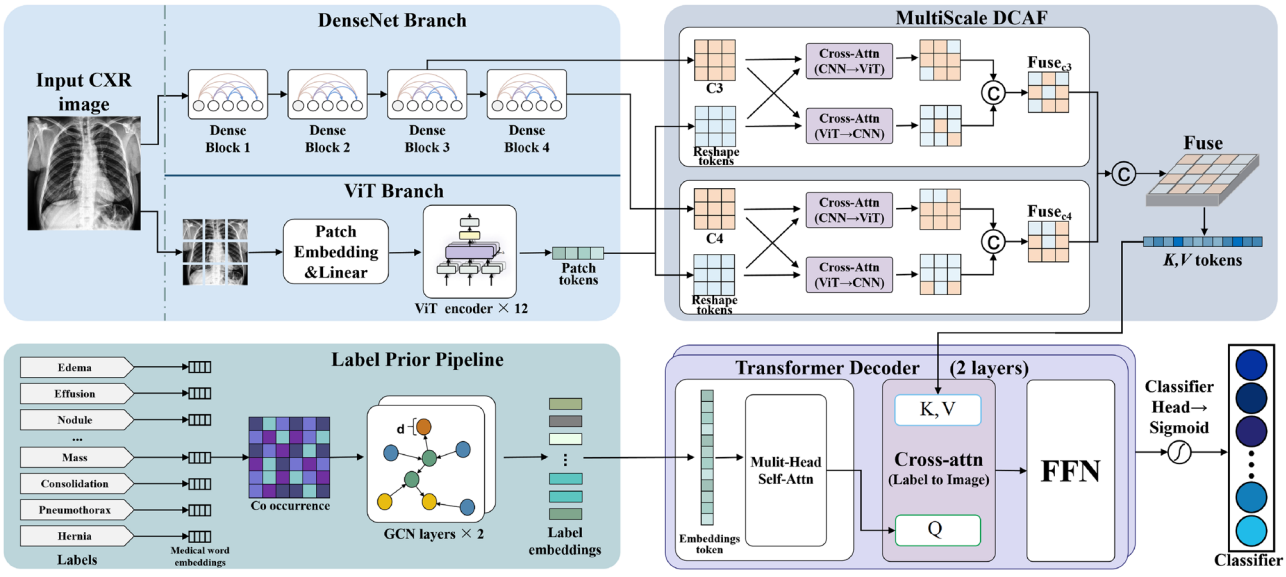


Fig. 1. Overall architecture of the proposed framework. The model uses DenseNet-121 and ViT as heterogeneous visual encoders, applies DCAF to obtain the fused visual memory, initializes label queries with GCN-refined label embeddings, and predicts a single logits matrix Z for multi-label classification. The circled "C" denotes feature concatenation, and the sigmoid layer converts logits into label probabilities.

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representation and contextual interaction; it does not imply that the ViT branch is specialized to a single spatial range.

Multi-scale bidirectional dual cross-attention fusion

Rationale for using C_3 and C_4

The choice of fusion stages needs to balance spatial resolution and class-discriminative semantics. Earlier DenseNet stages such as C_1 and C_2 retain finer spatial layouts, but their features may be dominated by low-level edge and texture responses and may contain limited disease-discriminative semantics for image-level prediction. Deeper stages provide more abstract semantics, but excessive downsampling may weaken subtle visual evidence. We therefore use C_3 and C_4 as the default fusion stages, allowing DCAF to interact with the ViT feature grid at two complementary levels of the convolutional hierarchy.

Feature projection and scale alignment

Before cross-representation interaction, DenseNet and ViT features are projected to a shared embedding dimension *d*. For each scale $s \in \{3, 4\}$, the DenseNet feature map is projected by a pointwise mapping ϕ_s:

$$C'_s = \phi_s(C_s), \qquad C'_s \in \mathbb{R}^{B\times d\times H_s\times W_s}. \tag{4}$$

The ViT feature grid is first projected to the same channel dimension and then resized to the spatial resolution of C'_s:

$$T_s = \mathrm{Resize}\left(\phi_v^{(s)}(T); H_s, W_s\right), \qquad T_s \in \mathbb{R}^{B\times d\times H_s\times W_s}. \tag{5}$$

Here, $\phi_v^{(s)}$ denotes the scale-specific projection applied to the ViT feature grid for alignment with the DenseNet feature map at stage *s*. The aligned pair (C'_s, T_s) is then used for bidirectional cross-attention at scale *s*.

Cross-attention operation

Figure 2 illustrates the basic cross-attention operation used in DCAF. Given a query sequence $Q \in \mathbb{R}^{N_Q\times d}$, a key sequence $K \in \mathbb{R}^{N_K\times d}$, and a value sequence $V \in \mathbb{R}^{N_K\times d}$, the scaled dot-product cross-attention is defined as

$$\mathrm{CrossAttn}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \tag{6}$$

where *d* is the feature dimension, $K^\top$ denotes matrix transposition, and the softmax is computed along the key dimension. For simplicity, linear query, key, and value projections are absorbed into the feature projection operators used before the attention operation.

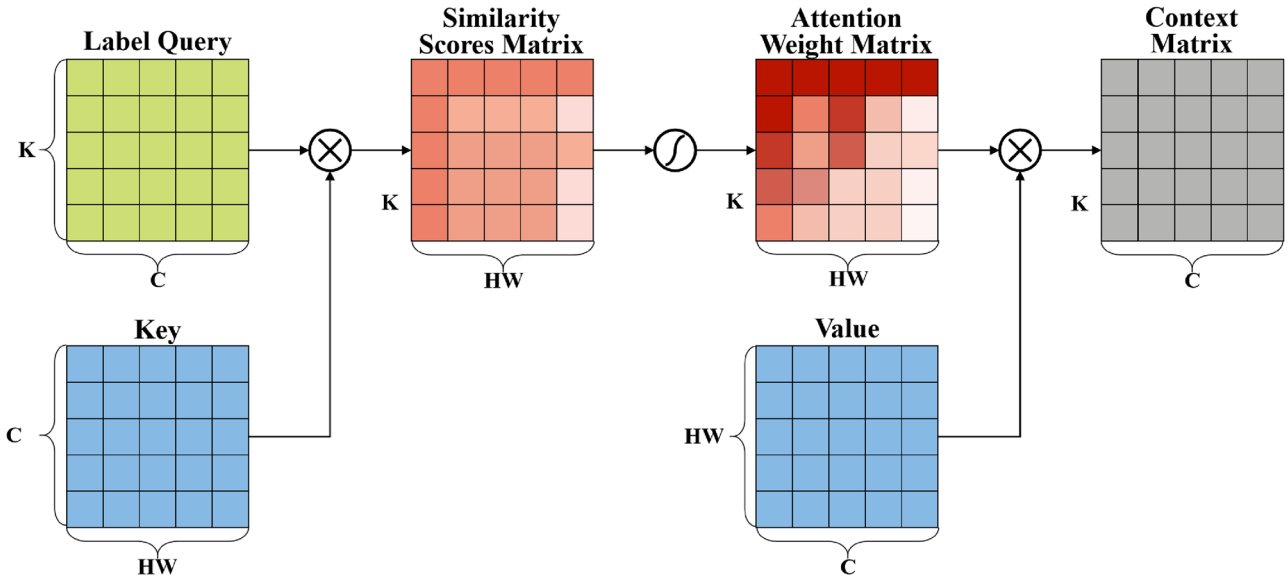


Fig. 2. Principle of the cross-attention operation used in DCAF. The query feature interacts with the key feature through scaled dot-product attention to generate normalized attention weights, which are then applied to the value feature to aggregate source information. In DCAF, this operation is performed in both directions between DenseNet convolutional feature maps and ViT token features.

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Bidirectional cross-representation fusion

For each scale $s \in \{3, 4\}$, the aligned maps C'_s and T_s are flattened into sequences along the spatial dimension:

$$C_s^\flat, T_s^\flat \in \mathbb{R}^{B \times N_s \times d}, \qquad N_s = H_s W_s. \tag{7}$$

Here, the superscript $\flat$ denotes flattening a two-dimensional feature map into a token sequence. The convolutional representation is updated by attending to the ViT tokens, and the ViT representation is updated by attending to the convolutional tokens:

$$\tilde{C}_s^\flat = C_s^\flat + \rho_c^{(s)}\left(\mathrm{CrossAttn}(C_s^\flat, T_s^\flat, T_s^\flat)\right), \tag{8}$$

$$\tilde{T}_s^\flat = T_s^\flat + \rho_t^{(s)}\left(\mathrm{CrossAttn}(T_s^\flat, C_s^\flat, C_s^\flat)\right), \tag{9}$$

where $\rho_c^{(s)}$ and $\rho_t^{(s)}$ denote learnable output projections. The tilde symbol denotes the cross-attention-refined feature, and the superscript (s) denotes scale-specific parameters rather than exponentiation. The updated sequences are reshaped back to two-dimensional feature maps, denoted by $\tilde{C}_s$ and $\tilde{T}_s$. They are then concatenated and compressed with a pointwise projection ψ_s:

$$F_s = \psi_s\left([\tilde{C}_s; \tilde{T}_s]\right), \qquad F_s \in \mathbb{R}^{B \times d \times H_s \times W_s}. \tag{10}$$

Here, $[\cdot;\cdot]$ denotes channel-wise concatenation and ψ_s denotes a pointwise projection at scale s. To aggregate information across scales, F_4 is upsampled to the resolution of F_3, and the two scale-specific features are concatenated and projected:

$$F = \psi\left([F_3; \mathrm{Up}(F_4)]\right), \qquad F \in \mathbb{R}^{B \times d \times H_f \times W_f}. \tag{11}$$

Here, $\mathrm{Up}(\cdot)$ denotes upsampling to the spatial resolution of F_3, and ψ denotes the pointwise projection used for multi-scale aggregation. The final fused feature map F is used as the visual memory for the decoder. Figure 3 summarizes the complete DCAF workflow, including scale alignment, bidirectional cross-attention refinement, and multi-scale feature aggregation.

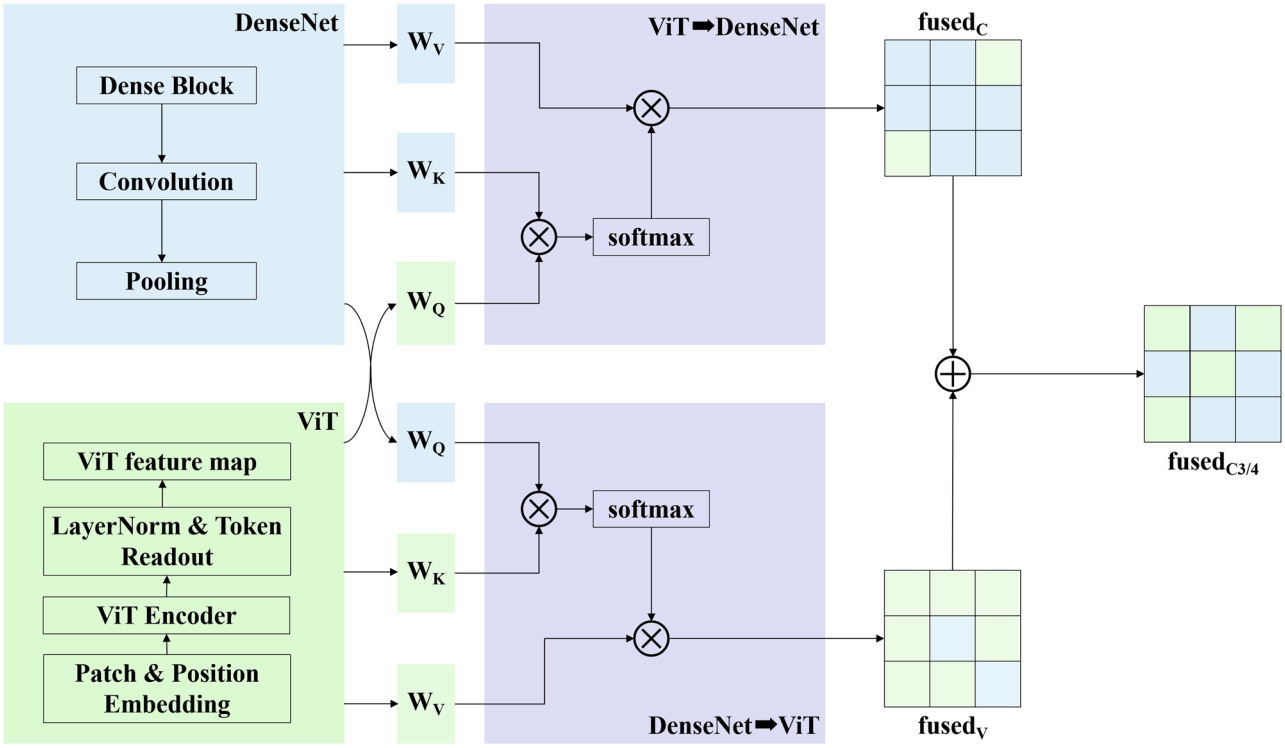


Fig. 3. Structure of the DCAF module. DenseNet feature maps C'_s and ViT feature maps T_s are aligned at scale $s \in \{3, 4\}$, flattened into token sequences, and refined through bidirectional cross-attention. The updated features are concatenated and projected to obtain F_s, and the multi-scale outputs are aggregated to form the final visual memory feature map F.

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ML-GCN-style label graph construction

Conditional co-occurrence matrix

Let $Y_{\mathrm{tr}} \in \{0,1\}^{N_{\mathrm{tr}} \times L}$ denote the binary label matrix of the training split. Here, $(Y_{\mathrm{tr}})_{ni} = 1$ indicates that the *n*-th training image is annotated with the *i*-th disease label, and $(Y_{\mathrm{tr}})_{ni} = 0$ otherwise. The co-occurrence count between labels *i* and *j* is

$$M_{ij} = \sum_{n=1}^{N_{\mathrm{tr}}} (Y_{\mathrm{tr}})_{ni} (Y_{\mathrm{tr}})_{nj}, \tag{12}$$

and the number of training samples containing label *i* is

$$N_i = \sum_{n=1}^{N_{\mathrm{tr}}} (Y_{\mathrm{tr}})_{ni}. \tag{13}$$

The directed conditional co-occurrence probability is then computed as

$$P_{ij} = P(L_j \mid L_i) = \frac{M_{ij}}{N_i + \varepsilon}, \tag{14}$$

where ε is a small constant for numerical stability. Here, L_i and L_j denote the *i*-th and *j*-th disease labels, respectively. Because $P(L_j \mid L_i)$ and $P(L_i \mid L_j)$ are generally different, *P* is a directed relation matrix.

Thresholded correlation matrix

Following the ML-GCN-style graph construction [7], we retain only sufficiently strong conditional relations using a fixed threshold $\tau = 0.4$:

$$\mathcal{B}_{ij} = \begin{cases} 1, & P_{ij} \geq \tau, \\ 0, & P_{ij} < \tau, \end{cases} \tag{15}$$

where $\mathcal{B} \in \{0,1\}^{L \times L}$ denotes the thresholded binary correlation matrix. No validation or test labels are used to construct *P* or $\mathcal{B}$.

Re-weighted correlation matrix

To control the strength of label-neighbor propagation and preserve self-information, we use the ML-GCN re-weighting scheme [7]. Let $d_i^{\mathcal{B}} = \sum_{k \neq i} \mathcal{B}_{ik}$ denote the number of off-diagonal neighbors of label *i*, where *k* is a label index. The final graph correlation matrix $A \in \mathbb{R}^{L \times L}$ is defined as

$$A_{ij} = \begin{cases} \dfrac{p}{d_i^{\mathcal{B}}}, & i \neq j,\ \mathcal{B}_{ij} = 1,\ d_i^{\mathcal{B}} > 0, \\ 1 - p, & i = j,\ d_i^{\mathcal{B}} > 0, \\ 1, & i = j,\ d_i^{\mathcal{B}} = 0, \\ 0, & \text{otherwise}, \end{cases} \tag{16}$$

where $p = 0.2$ controls the total weight assigned to neighboring labels. This design preserves label-specific information while allowing statistically correlated labels to exchange information.

Semantic node initialization and GCN propagation

Semantic information is introduced through node features rather than through semantic-similarity-based graph edges. Each disease label name is represented using GloVe 840B word vectors. For a multi-word label, the initial label embedding is computed by averaging the word vectors of its constituent words. The resulting initial node-feature matrix is

$$H^{(0)} \in \mathbb{R}^{L \times d_0}, \tag{17}$$

where $d_0 = 300$ for GloVe embeddings. A two-layer GCN propagates information over the re-weighted label graph:

$$H^{(1)} = \delta(AH^{(0)}W_0), \tag{18}$$

$$H^{(2)} = \delta(AH^{(1)}W_1), \tag{19}$$

where W_0 and W_1 are learnable parameters and $\delta(\cdot)$ denotes the nonlinear activation in the GCN layers. The final label embeddings are denoted by

$$H^{(2)} = [h_1^\top; \ldots; h_L^\top] \in \mathbb{R}^{L \times d}. \tag{20}$$

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Here, h_i denotes the GCN-refined embedding of label *i*. These embeddings are used to initialize the Transformer decoder queries.

Handling of the No Finding label
The *No Finding* label denotes the absence of annotated abnormalities and is not treated as a conventional disease category. For ChestX-ray14, the reported Mean AUC is computed over the 14 disease labels, and *No Finding* is not included in the disease-label graph used for label reasoning. For CheXpert, when *No Finding* is included in the 14-label setting, we avoid using its co-occurrence with disease labels as disease-promoting graph edges. Instead, *No Finding* is handled as a healthy-state indicator with its own label representation. The final prediction is still determined by label-conditioned visual evidence retrieved from the fused visual memory, rather than being forced by graph priors alone.

Graph-guided transformer decoder

Visual memory
The fused feature map $F \in \mathbb{R}^{B\times d\times H_f\times W_f}$ is flattened into a sequence of visual memory tokens:

$$M = \mathrm{Flatten}(F) \in \mathbb{R}^{B\times N\times d}, \qquad N = H_f W_f. \tag{21}$$

This memory contains the multi-scale cross-representation visual evidence produced by DCAF.

Label query initialization
The GCN-refined label embeddings $H^{(2)} \in \mathbb{R}^{L\times d}$ are expanded along the batch dimension to initialize the decoder queries:

$$Q^{(0)} = \mathrm{Expand}(H^{(2)}) \in \mathbb{R}^{B\times L\times d}. \tag{22}$$

Each query corresponds to one disease label and carries both co-occurrence-based structural information and semantic label initialization.

Label-to-image evidence retrieval
The Transformer decoder updates the label queries through stacked decoder layers. At each layer, query self-attention models interactions among label queries, while cross-attention retrieves label-specific visual evidence from *M*. For decoder layer *l*, this process can be summarized as

$$\bar{Q}^{(l)} = \mathrm{SelfAttn}(Q^{(l)}, Q^{(l)}, Q^{(l)}), \tag{23}$$

$$Q^{(l+1)} = \mathrm{CrossAttn}(\bar{Q}^{(l)}, M, M), \tag{24}$$

Here, $\bar{Q}^{(l)}$ denotes the self-attention-updated intermediate query representation at decoder layer *l*; residual connections, layer normalization, and a feed-forward network are applied in the standard Transformer-decoder manner after each sub-layer. After the final decoder layer, the output query representations are denoted by

$$Q_{\mathrm{out}} \in \mathbb{R}^{B\times L\times d}, \qquad Q_{\mathrm{out}}[b,i,:] = q_{bi} \in \mathbb{R}^{d}. \tag{25}$$

Multi-label prediction
A shared linear prediction head maps each label-specific decoder output to one scalar logit:

$$Z_{bi} = w_o^{\top} q_{bi} + b_o, \tag{26}$$

where q_{bi} is the decoder output corresponding to image *b* and label *i*, and w_o and b_o are the shared classifier weight and bias, respectively. The resulting logits form a single prediction matrix $Z \in \mathbb{R}^{B\times L}$. The multi-label prediction probabilities are obtained as $\hat{Y} = \sigma(Z)$.

Training objective

Single-logits composite loss
The model is optimized using a composite multi-label loss applied to the single logits matrix *Z*. No additional prediction branches are used in the training objective. The total loss is

$$\mathcal{L} = \lambda_{\mathrm{ASL}}\mathcal{L}_{\mathrm{ASL}}(Z,Y) + \lambda_{\mathrm{BCE}}\mathcal{L}_{\mathrm{BCE}}(Z,Y) + \lambda_{\mathrm{GSL}}\mathcal{L}_{\mathrm{GSL}}(Z) + \lambda_{\mathrm{AUC}}\mathcal{L}_{\mathrm{AUC}}(Z,Y). \tag{27}$$

Here, each $\lambda_{\cdot}$ denotes the corresponding loss weight. This formulation follows the implementation of the composite multi-label loss used in the released code. Labels with value -1, when present, are ignored through the loss mask.

Loss components
$\mathcal{L}_{\mathrm{ASL}}$ denotes asymmetric loss, which down-weights easy negative labels and helps address severe positive–negative imbalance in multi-label CXR classification. $\mathcal{L}_{\mathrm{BCE}}$ denotes binary cross-entropy with logits and ignore-label support, providing stable point-wise supervision for each disease label. $\mathcal{L}_{\mathrm{GSL}}$ is a graph-smoothing regularizer based on the label graph adjacency matrix; it softly encourages related labels to produce less

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contradictory probabilities without replacing image-based evidence retrieval. $\mathcal{L}_{\mathrm{AUC}}$ is a batch-wise pairwise ranking surrogate that encourages positive samples to receive higher scores than negative samples for each label, which is consistent with AUC-based evaluation. During inference, only Z is used, and the final probabilities are computed by sigmoid activation.

Implementation details

Input images are resized to 384×384. The DenseNet-121 branch is initialized with ImageNet-pretrained weights, and the ViT branch uses `vit_base_patch16_384` initialized from an MAE-pretrained checkpoint when specified. MAE pretraining is used as a representation initialization strategy and does not imply that the ViT branch is specialized to a single spatial range. The default DCAF setting uses C_3 and C_4 as fusion stages.

For label graph construction, all co-occurrence statistics are computed only from the training split. The conditional co-occurrence threshold is fixed at $\tau = 0.4$, the ML-GCN re-weighting parameter is fixed at $p = 0.2$, and the node features are initialized with GloVe 840B label-name embeddings. For CheXpert, the main experiments use U-Zeros, where uncertain labels are mapped to 0, and the supplementary U-Ones analysis maps uncertain labels to 1.

Computational efficiency is evaluated with 384×384 inputs. Parameter count and FLOPs are measured for a single forward pass. Inference latency and peak memory are measured under evaluation mode without gradient computation, with batch size 1 on a single NVIDIA RTX A6000 GPU unless otherwise specified.

Algorithm summary

Algorithm 1 summarizes the training pipeline.

Input: Training set $\mathscr{D} = \{(X, Y)\}$; DenseNet branch f_{CNN}; ViT branch f_{ViT}; DCAF module f_{DCAF}; GCN f_{GCN}; Transformer decoder f_{Dec}; prediction head g; graph parameters $\tau = 0.4$, $p = 0.2$; number of training epochs E.
Output: Learned model parameters Θ.
Initialize model parameters Θ
Compute P from training labels using conditional co-occurrence statistics
Threshold P with $\tau = 0.4$ to obtain $\mathscr{B}$
Apply ML-GCN re-weighting to $\mathscr{B}$ with $p = 0.2$ to obtain A
Initialize $H^{(0)}$ with GloVe 840B label-name embeddings
for epoch $= 1$ **to** E **do**
  for *mini-batch* $(X, Y) \sim \mathscr{D}$ **do**
    $C_3, C_4 \leftarrow f_{\mathrm{CNN}}(X)$
    $T \leftarrow f_{\mathrm{ViT}}(X)$
    $F \leftarrow f_{\mathrm{DCAF}}(C_3, C_4, T)$
    $M \leftarrow \mathrm{Flatten}(F)$
    $H^{(2)} \leftarrow f_{\mathrm{GCN}}(A, H^{(0)})$
    $Q^{(0)} \leftarrow \mathrm{Expand}(H^{(2)})$
    $Q_{\mathrm{out}} \leftarrow f_{\mathrm{Dec}}(Q^{(0)}, M)$
    $Z \leftarrow g(Q_{\mathrm{out}})$
    $\mathscr{L} \leftarrow \lambda_{\mathrm{ASL}}\mathscr{L}_{\mathrm{ASL}}(Z, Y) + \lambda_{\mathrm{BCE}}\mathscr{L}_{\mathrm{BCE}}(Z, Y) + \lambda_{\mathrm{GSL}}\mathscr{L}_{\mathrm{GSL}}(Z) + \lambda_{\mathrm{AUC}}\mathscr{L}_{\mathrm{AUC}}(Z, Y)$
    Back-propagate $\mathscr{L}$ and update Θ
  end
end
return Θ

Algorithm 1. Training pipeline of the proposed framework

Experiments

The experiments are organized in a layered manner to improve readability and to keep the evaluation logic explicit. We first describe the datasets, evaluation protocols, label co-occurrence statistics, and implementation details. We then report the main comparison results as averages over three independent runs (seeds 0, 1, and 2), followed by component-level and fine-grained ablation studies. Additional analyses examine the healthy-sample behavior and preliminary shared-label cross-dataset performance of the final model. Finally, we report computational efficiency and qualitative Grad-CAM visualizations to assess the practical cost and interpretability of the proposed framework.

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Label	Train positive (count)	Train share (%)	Test positive (count)	Test share (%)
Infiltration	14,204	24.46%	6,112	22.47%
Effusion	9089	15.65%	4,658	17.12%
Atelectasis	8531	14.69%	3,279	12.05%
Mass	5782	9.96%	1,748	6.43%
Nodule	5027	8.66%	1,623	5.97%
Consolidation	3091	5.32%	1,815	6.67%
Pneumothorax	2860	4.93%	2,665	9.80%
Pleural_Thickening	2421	4.17%	1,143	4.20%
Cardiomegaly	1746	3.01%	1,069	3.93%
Emphysema	1525	2.63%	1,093	4.02%
Edema	1441	2.48%	925	3.40%
Fibrosis	1291	2.22%	435	1.60%
Pneumonia	910	1.57%	555	2.04%
Hernia	149	0.26%	86	0.32%

Table 1. Statistics of ChestX-ray14 lesion labels in the training and test splits. The training split contains $N = 86{,}524$ images with 58,067 positive label occurrences. Images labeled No Finding are retained in the dataset but excluded from the 14-lesion statistics. The independent test split contains $N = 25{,}596$ images with 27,206 positive label occurrences.

Label	Positive (count)	Share (%)
Support devices	116,001	22.50%
Lung opacity	105,581	20.48%
Pleural effusion	86,187	16.72%
Edema	52,246	10.13%
Atelectasis	33,376	6.47%
Cardiomegaly	27,000	5.24%
No finding	22,381	4.34%
Pneumothorax	19,448	3.77%
Consolidation	14,783	2.87%
Enlarged cardiomediastinum	10,798	2.09%
Lung lesion	9186	1.78%
Fracture	9040	1.75%
Pneumonia	6039	1.17%
Pleural other	3523	0.68%

Table 2. Statistics of CheXpert label occurrences computed from the full train_val.csv file before the random 70%/10%/20% split. The dataset contains $N = 223{,}414$ samples with a total of 515,589 positive label occurrences. Only label value 1 is counted as positive, while -1 (uncertain) is treated as 0 (negative).

Datasets and evaluation protocols

We evaluate the proposed framework on two widely used benchmarks for multi-label chest X-ray (CXR) classification: ChestX-ray14 and CheXpert.

ChestX-ray14

The NIH ChestX-ray14 dataset contains 112,120 frontal chest radiographs collected from 32,717 patients and annotated with 14 thoracic disease labels[1]. We follow the official ChestX-ray14 evaluation protocol, using 86,524 images for model training and 25,596 images for testing. The original annotation system also includes a No Finding label; however, consistent with the evaluation protocol adopted in this work, the reported Mean AUC is averaged over the 14 disease labels only. Table 1 summarizes the label distribution in the training and test splits. In this table, "Share" denotes the proportion of positive occurrences of each label among all positive label occurrences in the corresponding split.

CheXpert

CheXpert is a large-scale CXR dataset that includes uncertainty labels[2]. The official CheXpert benchmark is based on a radiologist-annotated test protocol for five observations, whereas our study evaluates a 14-label setting. We therefore use the publicly released train_val.csv file and construct an internal evaluation split. Specifically, the full set is randomly shuffled with seed 0 and partitioned into 70%/10%/20% training/validation/test subsets.

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Uncertain labels are mapped to 0 following the U-Zeros policy in the main experiments. To assess the influence of uncertainty-label preprocessing, we additionally report a supplementary U-Ones setting, where uncertain labels are mapped to 1. In the CheXpert experiments, we adopt the 14-label prediction setting that includes No Finding. Table 2 summarizes the label distribution in the full train_val.csv file before splitting.

Both datasets exhibit pronounced long-tail label distributions. A small number of common findings account for most positive observations, whereas rare labels such as Hernia and Pleural Other occur only infrequently. This motivates the use of multi-scale visual fusion together with explicit modeling of label priors.

Evaluation metrics

We use the per-label area under the receiver operating characteristic curve (AUC) as the primary metric and report the arithmetic mean across labels (Mean AUC). For label c, the AUC is defined as

$$\mathrm{AUC}_c = \int_0^1 \mathrm{TPR}_c\left(\mathrm{FPR}_c^{-1}(x)\right)\, dx, \tag{28}$$

where TPR_c and FPR_c denote the true positive rate and false positive rate for label c, respectively. The Mean AUC is computed as

$$\text{Mean AUC} = \frac{1}{L}\sum_{c=1}^{L}\mathrm{AUC}_c, \tag{29}$$

where L is the number of evaluated labels. Because AUC is threshold-free, it is less sensitive to class imbalance than fixed-threshold accuracy metrics.

Co-occurrence statistics

To motivate explicit modeling of label dependencies, we compute label co-occurrence count matrices and visualize them as $\log(1+\text{count})$. We also compute conditional probability matrices $P(j|i)$ to characterize directed dependency strengths, as shown for ChestX-ray14 in Fig. 4 and for CheXpert in Fig. 5. In the actual label-graph construction, all graph statistics are recomputed from the training split only.

Implementation details

Training configuration

All experiments were conducted on a four-GPU server equipped with NVIDIA RTX A6000 GPUs (48 GB each). Input images were resized to 384×384. The model was trained for 30 epochs with a total batch size of 64. We optimized the network using SGD with momentum 0.9 and Nesterov acceleration. The initial learning rate was set to 0.006 and decayed by a factor of 0.2 at epochs $\{6, 12, 18\}$. Weight decay was set to 5×10^{-4} for the backbone and 1×10^{-4} for the newly added task-specific modules. The DenseNet branch was initialized with ImageNet-pretrained DenseNet-121 weights, whereas the Transformer branch used `vit_base_patch16_384` initialized from an MAE-pretrained checkpoint when specified. Before supervised training on ChestX-ray14 and CheXpert, we further pretrained the visual encoder on the MIMIC-CXR-JPG dataset to

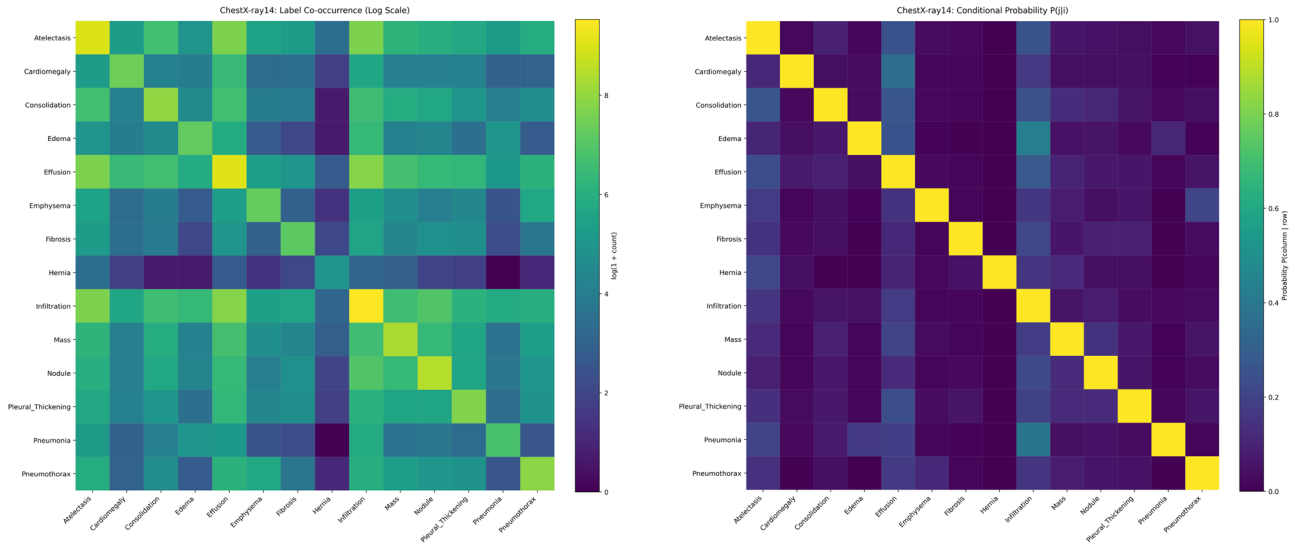
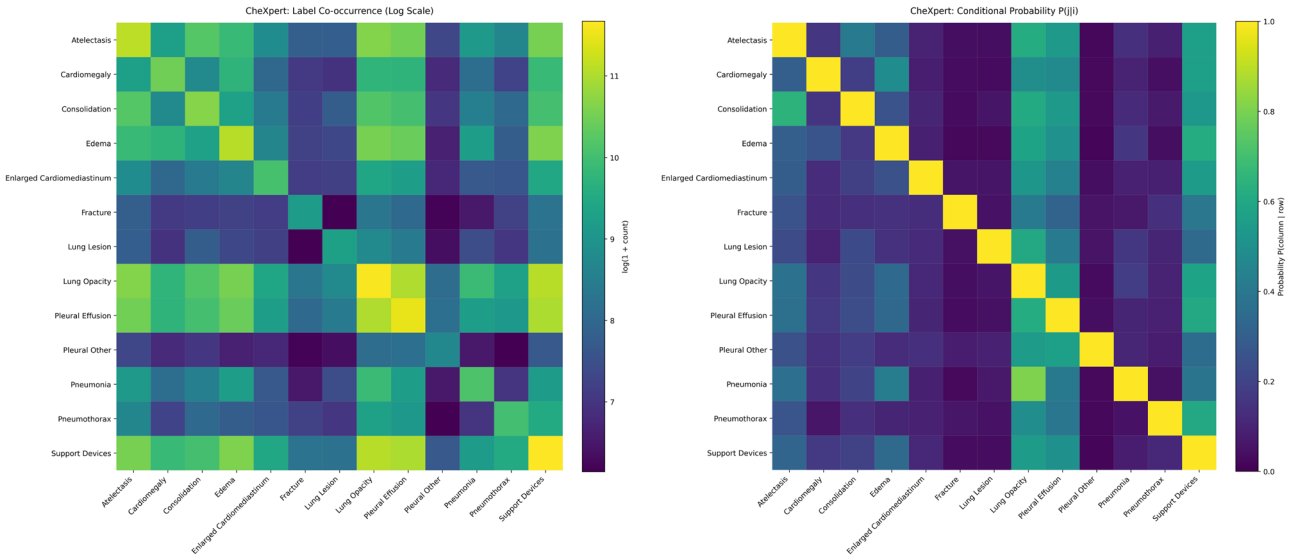


Fig. 4. ChestX-ray14 label co-occurrence statistics and conditional probability matrices. These visualizations are descriptive; the graph used for training is computed from the training split only.

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(a) (b)

Fig. 5. CheXpert label co-occurrence statistics and conditional probability matrices. These visualizations are descriptive; the graph used for training is computed from the training split only.

adapt it to the appearance of chest radiographs and common thoracic pathology patterns[35]. MAE and CXR-specific pretraining are treated as initialization strategies and do not imply that the ViT branch is specialized to a single spatial range. Afterward, the full network was fine-tuned end to end on each target benchmark. Data augmentation included random resized cropping, horizontal flipping, rotation within $\pm 15^{\circ}$, brightness/contrast jitter with factor 0.2, and random erasing. During validation and testing, we used deterministic resizing together with the same normalization pipeline.

Optimization objective
The model was trained with the single-logits composite objective defined in the Method section. In the reported implementation, the loss weights were $\lambda_{\mathrm{ASL}} = 1.0$, $\lambda_{\mathrm{BCE}} = 0.1$, $\lambda_{\mathrm{GSL}} = 0.03$, and $\lambda_{\mathrm{AUC}} = 0.06$. Only the single logits matrix Z is used in the training objective, and no additional prediction branches are optimized. The ASL hyperparameters were $\gamma_{\mathrm{neg}} = 4.0$, $\gamma_{\mathrm{pos}} = 1.0$, and $\mathrm{clip} = 0.05$. Labels with value -1 are ignored through the loss mask when applicable. Class-wise reweighting coefficients were estimated from the training data only and were used exclusively for loss reweighting during training; these coefficients were never computed on the final test split.

Label graph construction
For each dataset, label co-occurrence statistics were computed only from the training split. We followed the ML-GCN-style construction described in the Method section: directed conditional co-occurrence probabilities were thresholded with $\tau = 0.4$, the re-weighted correlation matrix used $p = 0.2$, and the initial node features were initialized from GloVe 840B label-name embeddings. For multi-word labels, the node feature was computed by averaging the embeddings of the constituent words. Semantic information is therefore introduced through node features rather than through semantic-similarity-based graph edges. No validation or test labels were used for graph construction.

Efficiency measurement
Computational efficiency was evaluated using input images resized to 384×384, consistent with the main experiments. Parameter count and FLOPs were computed for a single forward pass. Inference latency and peak GPU memory were measured on a single NVIDIA RTX A6000 GPU under evaluation mode without gradient computation. Unless otherwise specified, latency was measured with batch size 1 after warm-up iterations and averaged over repeated runs.

Training curves
Figure 6 shows the validation Mean AUC and loss curves during training. Under the final training protocol, the Mean AUC rises rapidly in the early stage and then stabilizes, while the validation loss decreases smoothly, indicating stable optimization of the hybrid encoder, fusion module, and decoder.

Main results

We first compare the proposed method with representative CNN-based, graph-based, Transformer-based, and hybrid methods on ChestX-ray14 and CheXpert. For the proposed method, we report the average AUC over three independent runs (seeds 0, 1, and 2) to provide a more stable estimate of classification performance.

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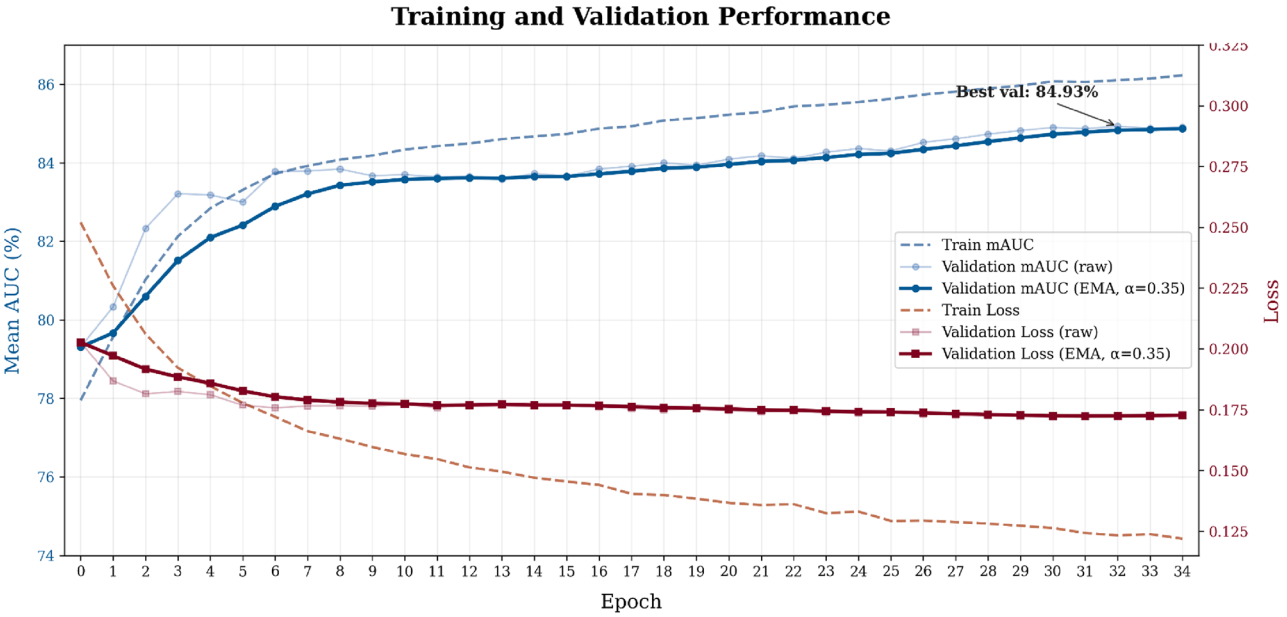


Fig. 6. Validation performance curves of the full model. The figure reports the validation Mean AUC and validation loss under the final training protocol.

Method	Ate	Con	Inf	Ptx	Ede	Emp	Fib	Eff	Pna	PT	Car	Nod	Mas	Her	Mean
Wang et al. [1]	0.716	0.708	0.609	0.806	0.835	0.815	0.769	0.784	0.633	0.708	0.807	0.671	0.706	0.767	0.738
Li et al. [36]	0.800	0.800	0.700	0.870	0.880	0.910	0.780	0.870	0.670	0.760	0.870	0.750	0.830	0.770	0.804
AG-CNN [11]	0.781	0.754	0.702	0.857	0.850	0.908	0.830	0.829	0.729	0.778	0.880	0.773	0.834	0.917	0.816
Jin et al. [15]	0.797	0.725	0.724	0.869	0.860	0.933	0.849	0.844	0.739	0.753	0.911	0.802	0.836	0.916	0.826
ImageGCN [30]	0.802	0.796	0.702	0.900	0.883	0.915	0.825	0.874	0.715	0.791	0.894	0.768	0.843	0.943	0.832
SynthEnsemble [6]	0.823	**0.810**	0.731	0.900	**0.902**	0.921	0.816	**0.882**	0.761	0.801	0.908	0.798	0.862	0.883	0.842
CXR-MLAGCPL [34]	0.799	**0.810**	0.717	0.873	0.901	0.882	0.799	0.877	0.763	0.783	0.904	0.728	0.810	0.941	0.828
Med-DGTN [37]	**0.829**	0.807	0.739	0.901	0.881	0.922	0.813	0.875	**0.781**	0.755	0.907	0.770	0.851	0.945	0.841
Ours	0.803	0.790	**0.742**	**0.903**	**0.902**	**0.936**	**0.856**	0.843	**0.781**	**0.812**	**0.913**	**0.806**	**0.863**	**0.948**	**0.849**

Table 3. Main comparison on ChestX-ray14. The table reports per-label AUC and Mean AUC following the official ChestX-ray14 evaluation protocol. The reported results of the proposed method are averaged over three independent runs (seeds 0, 1, and 2). *Ate* atelectasis, *Con* consolidation, *Inf* infiltration, *Ptx* pneumothorax, *Ede* edema, *Emp* emphysema, *Fib* fibrosis, *Eff* effusion, *Pna* pneumonia, *PT* pleural thickening, *Car* cardiomegaly, *Nod* nodule, *Mas* mass, *Her* hernia. Significant values are in bold.

Results on ChestX-ray14
As shown in Table 3, the proposed model achieves the highest Mean AUC among the listed methods on ChestX-ray14. The per-label results indicate clear gains for several subtle or clinically important findings, including Infiltration, Pneumothorax, Emphysema, Fibrosis, Pleural Thickening, Cardiomegaly, Nodule, Mass, and Hernia.

Results on CheXpert
Table 4 summarizes CheXpert performance and uncertainty-label sensitivity in a single view. U-Zeros is used as the default policy, while U-Ones is included as a supplementary setting to examine whether the conclusion depends strongly on how uncertain labels are mapped. For the proposed method, the reported values are also averaged over three independent runs (seeds 0, 1, and 2) under each uncertainty-label policy.

Ablation studies
Ablation studies are conducted on ChestX-ray14 and are organized from coarse to fine. We first evaluate the contribution of the major components, then examine the design of DCAF, the initialization of the ViT branch, and the construction of graph-guided label queries. Finally, a category-wise attribution analysis is used to relate module removal to disease-level performance changes.

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Method	NoFi	Enla	Card	Opac	Lesi	Edem	Cons	Pna	Atel	Ptx	PEff	POther	Frac	Supp	Mean
ML-GCN [7]	0.864	0.673	0.831	0.681	0.802	0.770	0.713	0.758	0.654	0.845	0.841	0.764	0.754	0.838	0.771
U_Zeros [2]	0.885	0.678	**0.865**	0.730	0.760	**0.853**	0.735	0.740	0.700	0.872	0.880	0.775	0.743	0.877	0.792
Swin-T [16]	0.845	0.611	0.800	0.706	0.720	0.791	0.674	0.719	0.679	0.761	0.834	0.675	0.682	0.774	0.734
LCHL_0s [31]	0.835	0.624	0.816	0.712	0.733	0.820	0.679	0.711	0.701	0.764	0.818	0.766	0.731	0.817	0.752
CXR×MLAGCPL_0s [34]	0.898	0.665	0.864	0.748	0.786	0.847	0.708	0.773	0.741	0.885	**0.890**	0.825	0.782	0.856	0.805
Ours (U-Zeros)	**0.901**	**0.682**	**0.865**	**0.760**	**0.811**	**0.853**	**0.750**	**0.775**	**0.744**	**0.886**	0.885	**0.830**	**0.786**	**0.879**	**0.815**
Ours (U-Ones)	0.892	0.674	0.858	0.752	0.803	0.846	0.742	0.768	0.738	0.878	0.879	0.823	0.779	0.872	0.807

Table 4. Main comparison and uncertainty-label analysis on CheXpert. The table reports representative published results together with our 14-label results under the internal 70%/10%/20% training/validation/testing partition of the publicly released `train_val.csv` file. Because published studies may use different split protocols, label subsets, and uncertainty-label policies, these values are intended as contextual reference rather than a strictly protocol-matched comparison. The reported results of the proposed method are averaged over three independent runs (seeds 0, 1, and 2) under the corresponding uncertainty-label policy. *NoFi* no finding, *Enla* enlarged cardiomediastinum, *Card* cardiomegaly, *Opac* lung opacity, *Lesi* lung lesion, *Edem* edema, *Cons* consolidation, *Pna* pneumonia, *Atel* atelectasis, *Ptx* pneumothorax, *PEff* pleural effusion, *POther* pleural other, *Frac* fracture, *Supp* support devices. Significant values are in bold.

Setting	Visual encoder	Classifier	Mean AUC (%)
ML-GCN baseline	ResNet	GCN classifier	76.1
DenseNet branch only	DenseNet	GCN classifier	82.0
ViT branch only	ViT	GCN classifier	82.3
ViT + DenseNet + DCAF	ViT + DenseNet + DCAF	GCN classifier	83.2
Full model	ViT + DenseNet + DCAF	Graph-guided cross-attention classifier	**84.9**

Table 5. Component-wise ablation study on ChestX-ray14 (mean AUC, %). Significant values are in bold.

Ablation group	Variant	Mean AUC (%)	Δ vs. group default
Fusion stage	$C_1 + C_2$	83.5	-1.4
	C_3 only	84.2	-0.7
	C_4 only	84.1	-0.8
	$C_3 + C_4$	84.9	–
Fusion mechanism	Concatenation	83.5	-1.4
	Weighted sum	83.7	-1.2
	Unidirectional DCAF (ViT-to-DenseNet)	84.2	-0.7
	Unidirectional DCAF (DenseNet-to-ViT)	84.1	-0.8
	Bidirectional DCAF	84.9	–

Table 6. Fine-grained ablation of the DCAF module on ChestX-ray14. The fusion-stage group evaluates where DenseNet and ViT features should interact, while the fusion-mechanism group evaluates how the aligned features should be fused. The default configuration is $C_3 + C_4$ with bidirectional DCAF.

Table 5 shows the contribution of the main components. Replacing the ResNet backbone in the ML-GCN baseline with stronger single-branch encoders already improves performance, while combining ViT and DenseNet through DCAF further increases the Mean AUC. The full model obtains the best result after replacing the GCN classifier with the graph-guided cross-attention decoder, suggesting that visual fusion and label-aware evidence retrieval provide complementary benefits.

DCAF module design

Table 6 provides a fine-grained analysis of the DCAF module from two perspectives: the fusion stage and the fusion mechanism. For the fusion-stage comparison, the $C_1 + C_2$ setting produces the largest performance drop, suggesting that early convolutional features, despite their higher spatial resolution, may contain limited disease-discriminative semantics for image-level CXR classification. Using either C_3 or C_4 alone improves the Mean AUC, but both variants remain inferior to the default $C_3 + C_4$ setting. This indicates that C_3 and C_4 provide complementary information: C_3 preserves relatively richer spatial structure, whereas C_4 provides stronger semantic abstraction.

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ViT initialization	Mean AUC (%)	Δ vs. MAE
Random initialization	82.9	-2.0
ImageNet supervised initialization	83.8	-1.1
MAE-pretrained initialization	84.9	–

Table 7. Effect of ViT initialization on ChestX-ray14. All variants use the same DenseNet branch, DCAF module, ML-GCN-style label graph, decoder, and training protocol; only ViT initialization is changed.

Label graph setting	Mean AUC (%)	Δ vs. default
Learnable label queries without GCN	83.8	-1.1
Binary matrix ($\tau = 0.4$) + GloVe node features	84.3	-0.6
Re-weighted matrix ($\tau = 0.4, p = 0.2$) + one-hot node features	84.1	-0.8
Re-weighted matrix ($\tau = 0.4, p = 0.2$) + GloVe node features (default)	84.9	–

Table 8. Ablation study of ML-GCN-style label graph construction on ChestX-ray14. The default setting uses thresholded conditional co-occurrence edges, ML-GCN re-weighting, and GloVe label-name embeddings as node features.

Category	Full model AUC	Δ w/o DCAF	Δ w/o graph-guided decoder	Primary interpretation
Nodule	0.806	-0.024	-0.011	Multi-scale visual fusion
Fibrosis	0.856	-0.020	-0.009	Multi-scale visual fusion
Effusion	0.843	-0.010	-0.021	Label correlation reasoning
Atelectasis	0.803	-0.012	-0.019	Label correlation reasoning
Infiltration	0.742	-0.008	-0.010	Ambiguous weak label
Consolidation	0.790	-0.009	-0.011	Ambiguous weak label

Table 9. Category-wise attribution analysis on ChestX-ray14. Δ w/o DCAF and Δ w/o graph-guided decoder denote the AUC drop after removing the corresponding component from the full model.

For the fusion-mechanism comparison, simple concatenation and weighted summation perform worse than attention-based fusion, indicating that static feature combination is insufficient for fully exploiting the complementary properties of DenseNet feature maps and ViT token representations. The two unidirectional DCAF variants improve the Mean AUC compared with static fusion, which supports the effectiveness of cross-representation attention. However, both unidirectional variants remain inferior to bidirectional DCAF. This suggests that mutual refinement between convolutional features and Transformer tokens is important: ViT tokens provide long-range relational cues for convolutional feature refinement, while DenseNet feature maps complement Transformer tokens with spatially organized visual evidence. Overall, these results support the use of $C_3 + C_4$ fusion with bidirectional DCAF as the default configuration.

ViT initialization
Table 7 clarifies the role of MAE initialization. Any gain from MAE pretraining should be interpreted as stronger token-level representation learning and improved optimization stability, not as evidence that the ViT branch functions as a purely global extractor.

Graph-guided label-query construction
Table 8 separates the roles of label-query initialization, co-occurrence-derived graph structure, ML-GCN re-weighting, and semantic node features. This design is aligned with the Method section, where graph edges are derived from conditional co-occurrence statistics and label-name embeddings are used only as node features.

Category-wise attribution
Table 9 is designed to explain non-uniform category-wise improvements. Larger drops after removing DCAF would support the role of multi-scale visual fusion for subtle findings such as *Nodule* and *Fibrosis*. Larger drops after removing the graph-guided decoder would support the role of label-prior-guided reasoning for clinically correlated findings such as *Effusion* and *Atelectasis*. Limited or fluctuating gains for diffuse categories should be interpreted cautiously because image-level labels may not fully capture their spatial and radiographic ambiguity.

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Method	No finding AUC	Healthy specificity	Diseased sensitivity	Mean AUC
Full model	0.901	0.866	0.823	0.815

Table 10. Analysis of *No Finding* and healthy-sample prediction of the final model on CheXpert. Healthy specificity evaluates whether healthy samples are correctly kept negative for disease labels, while diseased sensitivity evaluates whether images with at least one disease label are detected as abnormal.

Method	ChestX-ray14 → CheXpert	CheXpert → ChestX-ray14	Average
Full model	0.752	0.767	0.760

Table 11. Preliminary shared-label cross-dataset analysis of the final model on seven semantically matched labels between ChestX-ray14 and CheXpert. The model is trained on the source dataset and directly evaluated on the target dataset without target-domain fine-tuning. Mean AUC is computed over the seven shared labels only. These results are intended as complementary evidence rather than complete clinical external validation.

Method	Params (M)	FLOPs (G)	Latency (ms/img)	Memory (GB)	Mean AUC (%)
DenseNet branch only	8.1	11.5	18.4	1.3	82.0
ViT branch only	86.6	55.4	37.8	2.7	82.3
ViT + DenseNet + DCAF	102.4	72.8	58.6	4.6	83.2
Full model	108.9	75.3	64.2	4.9	84.9

Table 12. Computational efficiency comparison on ChestX-ray14. Params denotes trainable parameters, and FLOPs are computed for a single 384×384 input image. Latency and peak GPU memory are measured under evaluation mode with batch size 1 on a single NVIDIA RTX A6000 GPU to reflect single-image inference cost.

Additional analyses

The following analyses are complementary to the main comparison and ablation studies. They describe the behavior of the final model on healthy samples and under a preliminary shared-label cross-dataset setting, rather than serving as additional component ablations.

No finding and healthy-sample behavior

Table 10 provides a descriptive check of whether the final model maintains reasonable specificity on healthy samples while retaining sensitivity to images with at least one disease label. Because this analysis reports the final model only, it is intended to characterize healthy-sample behavior rather than to serve as a component-level ablation.

Preliminary shared-label cross-dataset analysis

To provide preliminary evidence of robustness under dataset shift, we conduct a shared-label cross-dataset analysis using the final model. Because ChestX-ray14 and CheXpert differ in label taxonomy, annotation policy, and uncertainty-label handling, we evaluate only seven labels with clear semantic correspondence: *Atelectasis*, *Cardiomegaly*, *Consolidation*, *Edema*, *Effusion*/*Pleural Effusion*, *Pneumonia*, and *Pneumothorax*. The model trained on the source dataset is directly evaluated on the target dataset without target-domain fine-tuning. This analysis is intended to characterize the final model under a controlled shared-label setting rather than to establish complete external clinical validation.

Table 11 reports the direct cross-dataset performance of the final model in both transfer directions. The results should be interpreted cautiously as complementary evidence, because the two datasets differ in population composition, imaging protocol, label definitions, annotation policy, and uncertainty-label handling.

Computational efficiency analysis

Table 12 evaluates whether the performance gain of the proposed framework is accompanied by acceptable computational overhead. Compared with the DenseNet branch only baseline, the full model increases the parameter count from 8.1M to 108.9M and the FLOPs from 11.5G to 75.3G, while improving Mean AUC from 82.0% to 84.9%. This indicates that the proposed framework introduces a substantial but expected computational cost due to the dual visual branches, DCAF, graph reasoning, and decoder-based evidence retrieval. Compared with the ViT + DenseNet + DCAF variant, the graph-guided decoder adds a relatively moderate overhead while improving Mean AUC from 83.2% to 84.9%, suggesting that label-prior-guided evidence retrieval provides a favorable performance gain relative to its additional cost.

Qualitative Grad-CAM visualization

Because the primary objective of this study is image-level multi-label classification rather than lesion detection or segmentation, Grad-CAM is used only as a qualitative interpretability tool. The visualization is designed to

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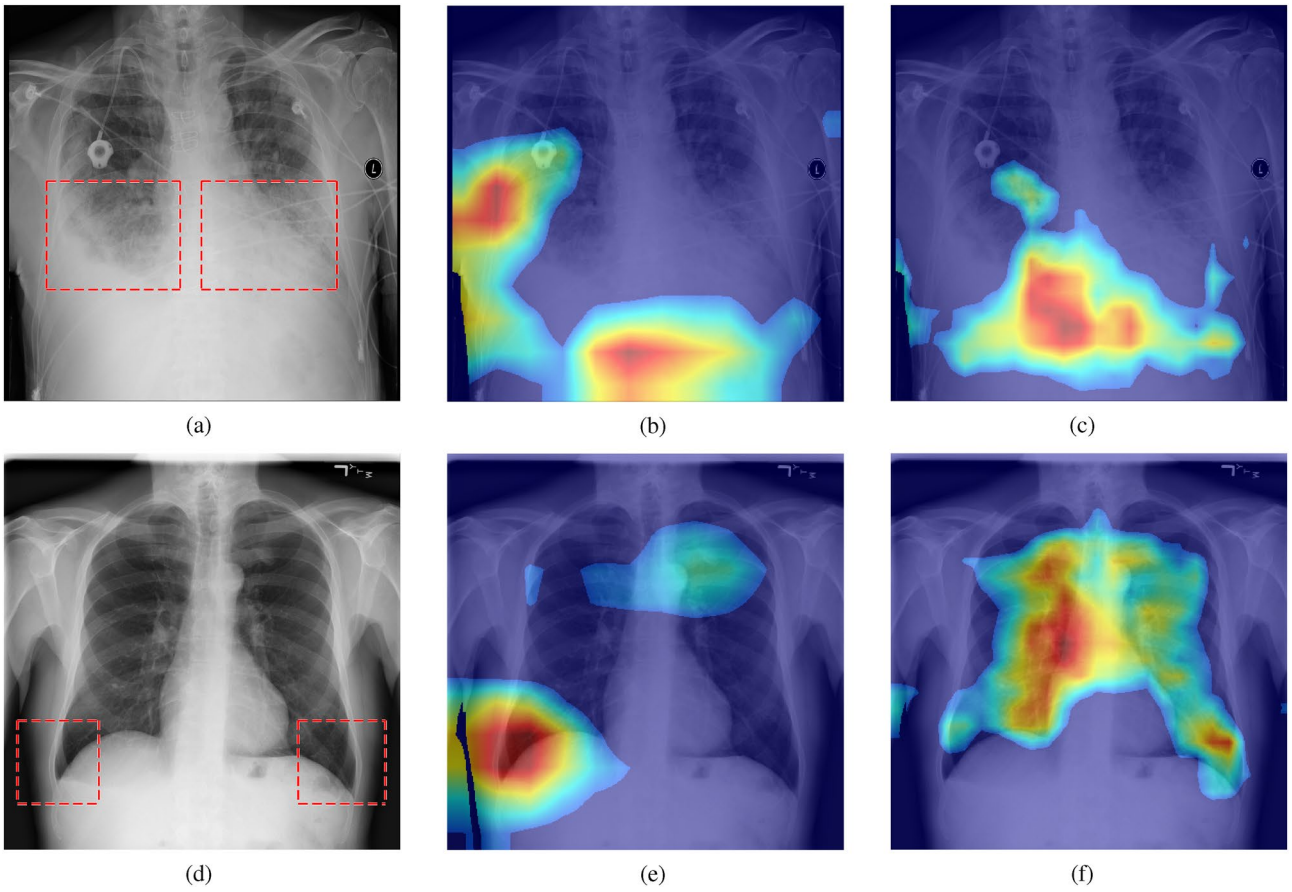


Fig. 7. Qualitative Grad-CAM visualization for two representative CXR cases. The first row shows a pneumonia case, and the second row shows a pleural effusion case. For each case, the leftmost image shows the input radiograph with manually indicated reference regions, followed by the Grad-CAM heatmap of the DenseNet-121 baseline and that of the proposed method. The thin dashed red rectangles denote approximate visually suspicious regions and are provided only as visual references for qualitative comparison. In the pneumonia case, the reference regions are placed around bilateral lower-lung suspicious opacities. In the pleural effusion case, the reference regions are placed around bilateral lower pleural and costophrenic-angle areas. These manually indicated regions are not ground-truth localization annotations and are not used for training, model selection, or quantitative evaluation.

examine whether the image regions emphasized by the model are broadly consistent with visually suspicious thoracic regions, rather than to provide a formal localization assessment.

Figure 7 shows two representative cases and compares the DenseNet-121 baseline with the proposed method. For each case, we present the original radiograph with manually indicated reference regions, the Grad-CAM map of the DenseNet-121 baseline, and the Grad-CAM map of the proposed method. In the leftmost image of each row, thin dashed red rectangles are manually added to indicate approximate visually suspicious reference regions for reader guidance. These reference regions are introduced only to facilitate qualitative comparison between the DenseNet-121 baseline and the proposed method. They are not lesion annotations, are not used during training or model selection, and are not used for quantitative localization evaluation.

In the pneumonia case, the manually indicated reference regions are located mainly in the bilateral lower lung fields, where suspicious parenchymal opacities can be visually observed. The DenseNet-121 baseline shows noticeable activation outside the most relevant lung regions, including regions close to external devices and image boundaries. In contrast, the proposed method produces a more concentrated response over the lung fields and shows better spatial agreement with the visually suspicious reference regions. This suggests that the multi-scale fused visual memory may help reduce attention to non-disease-related structures.

In the pleural effusion case, the manually indicated reference regions are placed around the bilateral lower pleural and costophrenic-angle areas, which are clinically relevant for pleural-effusion-like appearances. Compared with the baseline, the proposed method yields a broader and more coherent response over the lower thoracic and pleural-base regions. This behavior is consistent with the image-level prediction of pleural effusion, where the discriminative evidence may be distributed along the lower pleural spaces rather than confined to a small focal lesion.

Overall, these qualitative examples suggest that the proposed framework tends to activate regions that are more consistent with disease-relevant visual evidence than the DenseNet-121 baseline. However, because the

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datasets used in this study provide image-level labels rather than pixel-level lesion masks, these Grad-CAM results should be interpreted only as supportive visual evidence and not as proof of localization accuracy.

Conclusion

We presented a visual–semantic framework for image-level multi-label CXR classification that integrates heterogeneous visual representations with graph-guided label-aware decoding. The DenseNet–ViT encoder and DCAF module construct a multi-scale visual memory, while the ML-GCN-style label graph initializes decoder queries with co-occurrence-based structural priors and semantic label-name embeddings. Within the experiments reported here, the method achieves a Mean AUC of 0.849 on ChestX-ray14 following its official evaluation protocol and 0.815 on CheXpert using the internal 70%/10%/20% training/validation/testing partition described above. The multi-run averaged results, efficiency analysis, fine-grained ablations, healthy-sample behavior analysis, and preliminary shared-label cross-dataset analysis provide additional perspectives for characterizing robustness, computational cost, and cross-dataset behavior. These findings support the value of coupling cross-representation visual fusion with label-prior-guided evidence retrieval, while broader multi-center validation remains necessary before drawing strong conclusions about clinical deployment. Future work will also investigate how the current framework can interact with medical visual foundation models and biomedical vision–language pretraining, which may provide richer region-level priors and stronger semantic alignment under limited annotations and domain shift.

Data availability

The datasets analyzed during the current study are publicly available. The CheXpert dataset is available from Stanford University (https://stanfordmlgroup.github.io/competitions/chexpert/), and the ChestX-ray14 datase t is available from the National Institutes of Health (https://nihcc.app.box.com/v/ChestXray-NIHCC)[1,2]. The MIMIC-CXR-JPG dataset is available from the PhysioNet repository (https://physionet.org/content/mimic-cxr-jpg/), subject to credentialed access[35]. The public code repository provides split-generation scripts and pre-processing instructions; the original datasets are not redistributed and should be downloaded from their official sources according to the corresponding data-use agreements.

Code availability

The source code, training and evaluation scripts, and configuration files for the proposed method are publicly available at: https://github.com/sgk020204-a11y/DCAF-GCN-CXR. The released repository includes the implementation of the ViT–DenseNet visual encoder, the DCAF module, the ML-GCN-style label graph construction, the graph-guided Transformer decoder, the single-logits composite multi-label loss, training and evaluation scripts, configuration files, split-generation instructions, and ablation settings. The repository does not redistribute ChestX-ray14, CheXpert, or MIMIC-CXR-JPG; users should obtain these datasets from their official sources according to the corresponding data-use agreements.

Received: 30 March 2026; Accepted: 11 May 2026
Published online: 19 May 2026

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Author contributions

G.S. designed the study and collected and analyzed the data. Z.W.contributed to the study design, provided guidance on the data analysis, and critically reviewed the manuscript. Y.S.assisted with the data analysis. J.P. contributed to the interpretation of the results and critically reviewed the manuscript. L.S.provided guidance on the study design. F.F. contributed to the data analysis and interpretation.L.J.contributed to the study design, provided guidance on the data analysis, and wrote the manuscript.Each author has read and approved the final version of the manuscript and has agreed to be accountable for their contributions.

Funding

This work was supported by the Key Scientific Research Project of Higher Education Institutions of Anhui Province (Grant Nos. 2024AH051023, 2024AH051025), the Outstanding Young Backbone Talents of Anhui University Visiting and Training Abroad Project (Grant No. gxgwfx2019026), Anhui Provincial Quality Engineering Project (Grant No. 2023sx044), the Key Teaching Research Project of Anhui Province (Grant Nos. 2024jyxm0215, 2022jyxm851), the high-level talent introduction project of Anhui University of Chinese Medicine (Grant No. 2024rcyb025), the Key Teaching Research Project of Anhui University of Chinese Medicine (Grant No. 2023xjjy_zd028), and Quality Engineering Project of Anhui University of Chinese Medicine (Grant No. 2026xjzhkc-2).

Declarations

Competing Interests

The authors declare no competing interests.

Additional information

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